

WEEK 6: PROJECT SUMMARY REPORT.

Project Title: HealthConnect Clinic Experience Lab — Advanced Analytics & Decision Support

Track: Data Analytics Track.

1. Planned Objectives vs. Completed Work

- Planned: Upgrade the Week 5 baseline exploratory dashboard into an intermediate Decision Support System (v2.pbix), incorporate non-linear lead-time binning, implement cross-track feature handoffs, and validate operational KPIs.
- Completed: Successfully delivered HealthConnect Experience Lab Dashboard week 6 .pbix featuring an executive 4-KPI header, a 2D Risk Interaction Matrix, SMS-focused channel evaluation, a vertical sidebar container, and an exported feature summary matrix (ds_da_feature_matrix_v2.csv).

2. Improvements from Week 5

- Replaced fragmented "Risk Tier" labels with clear, operational lead-time and past-history groupings.
- Upgraded static charts into a dynamic, cross-filtering analytical environment where interacting with reminder channels or risk bins updates clinic metrics instantly.

3. Integration & Collaboration Summary

- Track Collaborated With: Data Science Track (Simulated Workflow / Handoff).
- Information Exchange: Provided validated non-linear lead-time bins (0–6d, 7–13d, 14–27d, 28+d) and historical no-show

probabilities; received predictive risk scoring thresholds and feature weighting structures.

- Project Impact: Converted descriptive historical reporting into structured feature inputs suitable for machine learning pipelines.

4. Key Analytical Findings & Development Outcomes

- The 30+ Day Critical Threshold: Patients booking more than 30 days in advance with a history of missed visits reach an attrition peak of 67.88%.
- Channel Efficiency: SMS reminders achieve the lowest baseline no-show rate (45.75%), confirming their priority for clinic intervention.
- Financial Impact: Total clinic revenue loss is quantified at \$121K based on a \$50 opportunity cost per unfulfilled slot.

5. Major Challenges & Decisions

- Challenge: Managing visual layer hierarchy and preventing background shapes from blocking slicer interactions in Power BI.
- Decision: Implemented a structured vertical sidebar container using the Selection Pane and locked background shapes to preserve dashboard usability.

6. Remaining Issues & Limitations

- Financial loss calculations rely on a static \$50 slot assumption without differentiating specialized medical departments.
- Environmental factors (such as public transit delays or regional weather) are currently omitted from the appointment logs.

7. Contribution & Week 7 Focus

- Overall Contribution: Provided the foundational operational metrics, interactive visual framework, and risk-segmentation logic for the multidisciplinary HealthConnect solution.

- Proposed Week 7 Focus: Execute end-to-end testing, validate visual interactions under simulated user queries, and prepare final documentation for stakeholder presentation